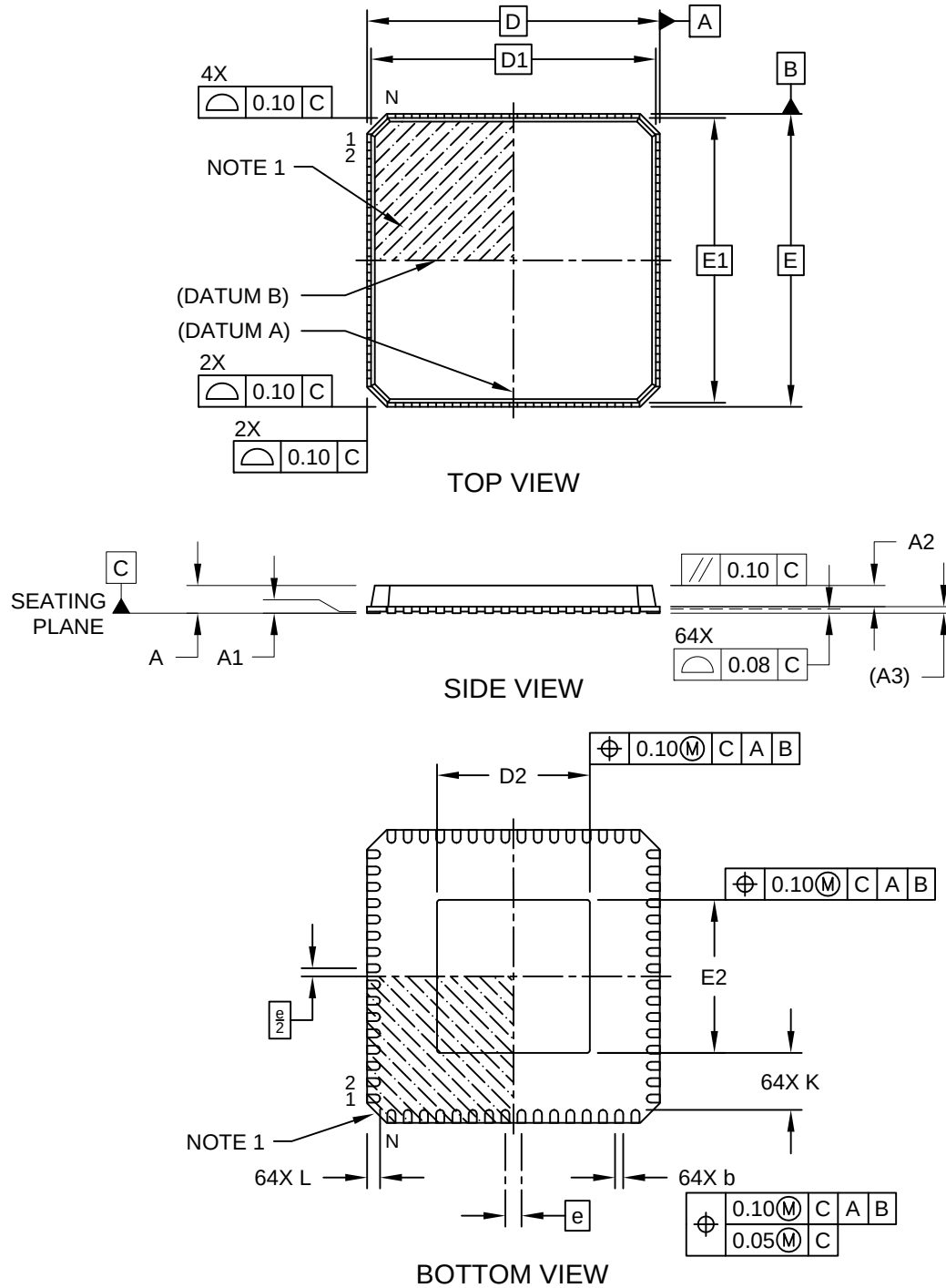


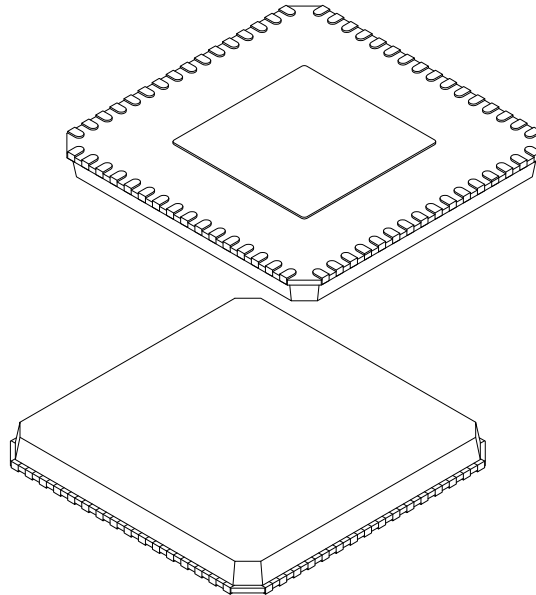
64-Lead Very Thin Plastic Quad Flat, No Lead Package (RG) - 9x9 mm Body [VQFN] **Punch Singulated**

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



64-Lead Very Thin Plastic Quad Flat, No Lead Package (RG) - 9x9 mm Body [VQFN] With 4.7x4.7 mm Exposed Pad; Punch Singulated

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	64		
Pitch	e	0.50 BSC		
Overall Height	A	0.80	0.85	1.00
Standoff	A1	0.00	0.02	0.05
Mold Cap Thickness	A2	0.65	-	0.80
Terminal Thickness	(A3)	0.20 REF		
Overall Length	D	9.00 BSC		
Mold Top Length	D1	8.75 BSC		
Exposed Pad Length	D2	4.60	4.70	4.80
Overall Width	E	9.00 BSC		
Mold Top Width	E1	8.75 BSC		
Exposed Pad Width	E2	4.60	4.70	4.80
Terminal Width	b	0.18	0.25	0.30
Terminal Length	L	0.30	0.40	0.50
Terminal-to-Exposed-Pad	K	1.55	-	-

Notes:

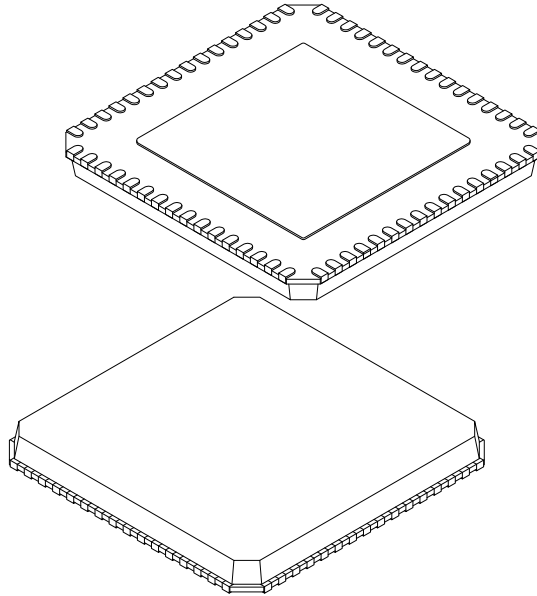
- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Package is saw singulated
- Dimensioning and tolerancing per ASME Y14.5M

BSC: Basic Dimension. Theoretically exact value shown without tolerances.

REF: Reference Dimension, usually without tolerance, for information purposes only.

**64-Lead Very Thin Plastic Quad Flat, No Lead Package (RG) - 9x9 mm Body [VQFN]
With 6.0x6.0 mm Exposed Pad; Punch Singulated**

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



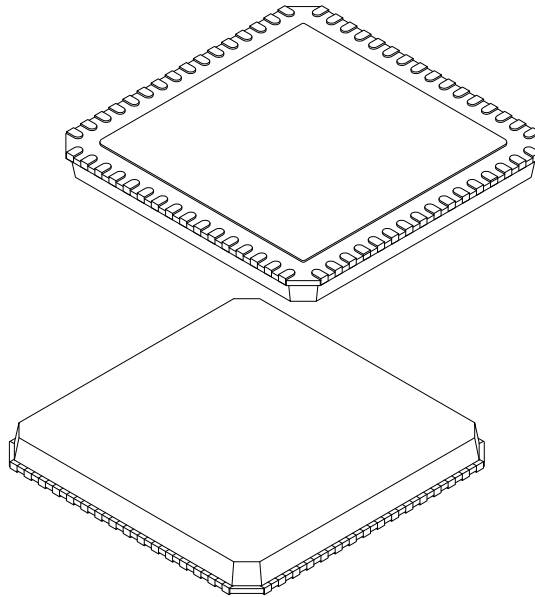
Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	64		
Pitch	e	0.50 BSC		
Overall Height	A	0.80	0.85	1.00
Standoff	A1	0.00	0.02	0.05
Mold Cap Thickness	A2	0.65	-	0.80
Terminal Thickness	(A3)	0.20 REF		
Overall Length	D	9.00 BSC		
Mold Top Length	D1	8.75 BSC		
Exposed Pad Length	D2	5.90	6.00	6.10
Overall Width	E	9.00 BSC		
Mold Top Width	E1	8.75 BSC		
Exposed Pad Width	E2	5.90	6.00	6.10
Terminal Width	b	0.18	0.25	0.30
Terminal Length	L	0.30	0.40	0.50
Terminal-to-Exposed-Pad	K	0.90	-	-

Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Package is saw singulated
- Dimensioning and tolerancing per ASME Y14.5M
 - BSC: Basic Dimension. Theoretically exact value shown without tolerances.
 - REF: Reference Dimension, usually without tolerance, for information purposes only.

**64-Lead Very Thin Plastic Quad Flat, No Lead Package (RG) - 9x9 mm Body [VQFN]
With 7.3x7.3 mm Exposed Pad; Punch Singulated**

Note: For the most current package drawings, please see the Microchip Packaging Specification located at <http://www.microchip.com/packaging>



Units		MILLIMETERS		
Dimension Limits		MIN	NOM	MAX
Number of Terminals	N	64		
Pitch	e	0.50 BSC		
Overall Height	A	0.80	0.85	1.00
Standoff	A1	0.00	0.02	0.05
Mold Cap Thickness	A2	0.65	-	0.80
Terminal Thickness	(A3)	0.20 REF		
Overall Length	D	9.00 BSC		
Mold Top Length	D1	8.75 BSC		
Exposed Pad Length	D2	7.20	7.30	7.40
Overall Width	E	9.00 BSC		
Mold Top Width	E1	8.75 BSC		
Exposed Pad Width	E2	7.20	7.30	7.40
Terminal Width	b	0.18	0.25	0.30
Terminal Length	L	0.30	0.40	0.50
Terminal-to-Exposed-Pad	K	1.55	-	-

Notes:

- Pin 1 visual index feature may vary, but must be located within the hatched area.
- Package is saw singulated
- Dimensioning and tolerancing per ASME Y14.5M
 - BSC: Basic Dimension. Theoretically exact value shown without tolerances.
 - REF: Reference Dimension, usually without tolerance, for information purposes only.